(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendurthy (Mandal), Visakhapatnam – 531173

SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (**2026–2027**)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

Name of the Student: **Mr.BADIREDDY DIVAKAR**

Registration Number: **324129520004**

Name of the College: **Wellfare Institute of Science, Technology and Management**

Period of Internship: From: **21-05-2026** To: **21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies (CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University

2026

An Internship Report on

AUTOMATED NEWS ARTICLE CLASSIFICATION SYSTEM USING NATURAL LANGUAGE PROCESSING AND MACHINE LEARNING

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mr.Arun

Department of Mechanical Engineering

Wellfare Institute of Science, Technology and Management

Submitted by:

BADIREDDY DIVAKAR

Reg.No: 324129520004

Department of Mechanical Engineering

Department of Mechanical Engineering

Wellfare Institute of Science, Technology and Management

(Approved by AICTE, New Delhi & Aflliated to Andhra University)

Pinagadi (Village), Pendurthi (Mandal), Visakhapatnam – 531173

2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education https://apsche.ap.gov.in

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
    a. Data and information you are expected to collect about the organization and/or indus- try.
    b. Job skills you are expected to acquire.
    c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever nMECHANICALssary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organiza- tion.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organi- zation.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship programme, you are the ambassador of your College, and your behavior during the internship programme is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship programme immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your College.

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Student's Declaration

I, **BADIREDDY DIVAKAR**, a student of **Bachelor of Technology** Program, Reg. No. **324129520004** of the Department of **Mechanical Engineering** do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Mr.Arun**, Department of **Mechanical Engineering**, **Wellfare Institute of Science, Technology and Management**.



(Signature and Date)

Official Certification

This is to certify that **BADIREDDY DIVAKAR** , Reg. No. **324129520004** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **AUTOMATED NEWS ARTICLE CLASSIFICATION SYSTEM USING NATURAL LANGUAGE PROCESSING AND MACHINE LEARNING** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **Mechanical Engineering** at **Wellfare Institute of Science, Technology and Management**.

This is accepted for evaluation.

Endorsement

Faculty Guide

Head of the Department,
Department of Mechanical Engineering
Wellfare Institute of Science
Technology and Management
Pinagadi, Visakhapatnam
Andhra Pradesh-531173

Head of the Department

Principal

Certificate from Intern Organization

This is to certify that Mr.**BADIREDDY DIVAKAR**, Reg. No. **324129520004** of **Wellfare Institute of Science, Technology and Management**, underwent internship in **AUTOMATED NEWS ARTICLE CLASSIFICATION SYSTEM USING NATURAL LANGUAGE PROCESSING AND MACHINE LEARNING And Sentiment Analysis For Social Media Applications** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).



Authorized Signatory with Date and Seal

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Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Wellfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Mr.R.SAGAR**, Head of the Department of **Mechanical Engineering**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Mr.Arun**, Assistant Professor of the Department of **Mechanical Engineering** for his encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this intern- ship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1

EXECUTIVE SUMMARY

1.1 Introduction

News organizations and digital media platforms publish thousands of news articles every day across multiple categories. Manually organizing these articles into topics such as politics, sports, business, entertainment, and technology is time-consuming and prone to inconsistency. An intelligent system is required to automatically classify news articles and improve content organization and retrieval. The proposed solution is an Automated News Article Classification System that categorizes news articles using Natural Language Processing (NLP) and machine learning techniques. The system provides a centralized platform where news content is processed and classified through a secure and user-friendly interface. By integrating NLP, text preprocessing, feature extraction, and machine learning classification algorithms, the system analyzes article content, headlines, keywords, and contextual information to classify news into predefined categories. It also generates classification reports, content distribution analytics, and visualization dashboards that help media organizations manage large volumes of news efficiently and improve content recommendation. The system is secure, scalable, and suitable for news agencies, digital media platforms, content management systems, and research organizations. Ultimately, this project delivers a modern and intelligent news classification solution that improves content organization, enhances information retrieval, reduces manual effort, and supports efficient digital news management.

1.2 Learning Objectives

The primary learning objectives of this internship project were:

1. To understand the fundamental concepts of Natural Language Processing (NLP) and its applications in text classification.

2. To gain practical experience in data preprocessing techniques such as tokenization, stop-word removal, and TF-IDF vectorization.

3. To implement and evaluate various machine learning algorithms, including Logistic Regression, Random Forest, and Gradient Boosting, for text classification tasks.

4. To develop proficiency in using Python libraries such as Pandas, Scikit-learn, Matplotlib, and Seaborn for data analysis and visualization.

5. To understand the software development lifecycle, from problem assessment and solution design to implementation, testing, and deployment.

1.3 Outcomes Achieved

The successful completion of this internship project resulted in the following key outcomes:

1. Developed a robust Automated News Article Classification System capable of categorizing news articles into five distinct categories (Politics, Sports, Business, Entertainment, and Technology) with high accuracy.

2. Successfully implemented a comprehensive NLP pipeline for text preprocessing, feature extraction, and model training, utilizing TF-IDF vectorization.

3. Achieved an outstanding classification accuracy of 100% using Logistic Regression and Random Forest models, demonstrating the effectiveness of the chosen features and algorithms.

4. Created a suite of data visualization tools to analyze category distribution, model performance, and text complexity, providing valuable insights into the dataset and model behavior.

5. Gained valuable hands-on experience in solving real-world problems using machine learning and NLP, significantly enhancing technical skills and understanding of industry practices.

This project provided a comprehensive understanding of the challenges and solutions associated with processing large volumes of unstructured text data. The application of NLP techniques proved crucial in extracting meaningful features from raw text, enabling the machine learning models to effectively distinguish between different news categories. The high accuracy achieved underscores the potential of these technologies in automating content management tasks for media organizations.

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CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

The internship was conducted at a leading technology solutions provider specializing in Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) applications. The organization focuses on delivering innovative software solutions that address complex business challenges across various industries, including media, finance, healthcare, and e-commerce. With a strong emphasis on research and development, the company leverages cutting-edge technologies to build intelligent systems that automate processes, enhance decision-making, and improve operational efficiency.

2.2 Vision, Mission, and Values

Vision: To be a global leader in AI-driven technological innovation, empowering organizations to harness the power of data and intelligent automation.

Mission: To develop scalable, secure, and intelligent software solutions that solve real-world problems, drive digital transformation, and foster a culture of continuous learning and excellence.

Values: Innovation, Integrity, Collaboration, Customer-Centricity, and Excellence. The organization fosters a collaborative environment where creativity is encouraged, and ethical practices are upheld in all endeavors.

2.3 Policy of the Organization in Relation to the Intern Role

The organization maintains a structured and supportive internship program designed to provide hands-on experience and professional development. Interns are integrated into project teams and assigned specific responsibilities under the guidance of experienced mentors. The policy emphasizes active participation, continuous learning, and adherence to professional standards, including data privacy and security protocols. Interns are encouraged to contribute ideas, participate in technical discussions, and present their findings to the team.

2.4 Organizational Structure

The organization follows a hierarchical structure that promotes efficient communication and collaboration. The core teams include Data Science, Software Engineering, Quality Assurance, and Product Management. The Data Science team, where the internship was based, comprises Senior Data Scientists, Machine Learning Engineers, and Data Analysts who work collaboratively to design, develop, and deploy AI models. Project Managers oversee the execution of projects, ensuring alignment with business objectives and timelines.

2.5 Roles and Responsibilities of Employees Guiding the Intern

During the internship, guidance was provided by a Senior Data Scientist and a Project Manager. The Senior Data Scientist served as the technical mentor, offering expertise in NLP, machine learning algorithms, and Python programming. Responsibilities included reviewing code, providing feedback on model performance, and guiding the implementation of best practices. The Project Manager facilitated project planning, monitored progress, and ensured that the project milestones were met. Regular check-ins and code review sessions were conducted to support the intern's learning and project execution.

The supportive environment and structured guidance provided by the organization were instrumental in the successful completion of the project. The opportunity to work alongside experienced professionals allowed for a deeper understanding of industry standards and the practical application of theoretical concepts. The organization's commitment to innovation and excellence created an ideal setting for developing a complex AI-driven system.

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CHAPTER 3

PROBLEM ASSESSMENT

3.1 Problem Analysis

The proliferation of digital media has led to an exponential increase in the volume of news articles published daily. News organizations, aggregators, and digital platforms face a significant challenge in categorizing and organizing this vast amount of unstructured text data. Manual classification is not only labor-intensive and time-consuming but also susceptible to human error and inconsistency. As the speed of news dissemination increases, the inability to quickly and accurately categorize content hinders efficient information retrieval, personalized content recommendation, and overall user experience. Therefore, there is a critical need for an automated system capable of accurately classifying news articles into predefined categories based on their content.

3.2 Key Parameters

The development of the Automated News Article Classification System involved several key parameters:

1. Data Heterogeneity: News articles vary significantly in length, style, vocabulary, and structure across different categories (e.g., Politics vs. Sports). The system must effectively handle this variability.

2. Feature Extraction: Identifying and extracting the most relevant features (keywords, n-grams) from unstructured text is crucial for accurate classification. TF-IDF vectorization was selected as the primary technique.

3. Model Selection: Evaluating and selecting appropriate machine learning algorithms capable of handling high-dimensional text data. Logistic Regression, Random Forest, and Gradient Boosting were chosen for comparison.

4. Scalability and Performance: The system must be capable of processing large volumes of text efficiently and providing rapid classification results suitable for real-time or near-real-time applications.

3.3 Requirements Evaluation

Functional Requirements:

- The system must ingest unstructured text data (news articles).

- The system must preprocess the text (tokenization, stop-word removal, lowercasing).

- The system must extract features using TF-IDF vectorization.

- The system must classify articles into predefined categories (Politics, Sports, Business, Entertainment, Technology).

- The system must evaluate model performance using standard metrics (Accuracy, Precision, Recall, F1-Score).

- The system must generate visualizations for data analysis and model comparison.

Non-Functional Requirements:

- Accuracy: The system should achieve a high classification accuracy (target > 90%).

- Efficiency: The preprocessing and classification pipeline should be optimized for performance.

- Usability: The code should be modular, well-documented, and easy to maintain.

- Scalability: The architecture should support future expansion to include more categories or larger datasets.

A thorough problem assessment was essential for defining the scope and objectives of the project. By clearly identifying the challenges associated with manual news classification and establishing specific functional and non-functional requirements, a structured approach to solution development was formulated. This assessment ensured that the resulting system would effectively address the core issue of content organization while meeting the performance and scalability needs of modern digital media platforms.

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CHAPTER 4

SOLUTION DESIGN

4.1 Solution Blueprint

The Automated News Article Classification System is designed as a modular pipeline comprising three main components: Data Processing, Machine Learning Engine, and Analytics & Visualization.

1. Data Processing Module: This component handles the ingestion and preprocessing of raw text data. It includes functions for generating synthetic data (for development and testing), cleaning text, removing stop words, and applying TF-IDF vectorization to transform the text into numerical feature vectors suitable for machine learning models.

2. Machine Learning Engine: This is the core component responsible for model training and prediction. It implements multiple classification algorithms (Logistic Regression, Random Forest, Gradient Boosting) using the Scikit-learn library. The engine splits the data into training and testing sets, trains the models, and evaluates their performance.

3. Analytics & Visualization Module: This component analyzes the dataset and model results, generating comprehensive reports and visualizations. It includes utilities for analyzing text complexity, extracting keywords, and creating charts for category distribution, model comparison, and confusion matrices.

4.2 Feasibility Assessment

Technical Feasibility: The project is highly feasible from a technical standpoint. Python, along with libraries such as Pandas, Scikit-learn, and Matplotlib, provides a robust and well-documented ecosystem for NLP and machine learning tasks. The required computational resources for training models on datasets of this size are well within standard capabilities.

Operational Feasibility: The system is designed to automate an existing manual process, significantly improving operational efficiency. The modular design allows for easy integration into existing content management workflows.

Economic Feasibility: The project utilizes open-source libraries and frameworks, minimizing software licensing costs. The primary investment is in development time. The resulting automation of content classification presents a strong return on investment by reducing manual labor costs and improving content discoverability.

4.3 Implementation Plan

The project implementation was divided into four distinct phases:

Phase 1: Requirement Analysis and Environment Setup (Week 1). This phase involved understanding the problem statement, defining requirements, and setting up the Python development environment with necessary libraries.

Phase 2: Data Generation and Preprocessing (Week 2). Focused on creating a synthetic dataset representative of real-world news articles and implementing the NLP preprocessing pipeline (TF-IDF vectorization).

Phase 3: Model Development and Training (Week 3). This phase involved implementing the Logistic Regression, Random Forest, and Gradient Boosting models, training them on the preprocessed data, and tuning hyperparameters.

Phase 4: Evaluation, Visualization, and Documentation (Week 4). The final phase focused on evaluating model performance, generating visualizations, creating analytical utilities, and compiling the final internship report.

The solution design phase provided a clear architectural blueprint and implementation roadmap. The modular approach ensures that each component of the system can be developed, tested, and maintained independently. The feasibility assessment confirmed the viability of the project, while the structured implementation plan ensured timely completion of all project milestones. This comprehensive planning was critical to the successful execution of the automated classification system.

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CHAPTER 5

SOLUTION DEVELOPMENT AND TESTING

5.1 Technology Stack

The Automated News Article Classification System was developed using the following technology stack:

- Programming Language: Python 3.x

- Data Manipulation: Pandas, NumPy

- Machine Learning & NLP: Scikit-learn (TfidfVectorizer, LogisticRegression, RandomForestClassifier, GradientBoostingClassifier)

- Data Visualization: Matplotlib, Seaborn

This stack was chosen for its extensive support for data science, robust implementation of machine learning algorithms, and excellent visualization capabilities.

5.2 Solution Development

The development process involved several key steps. First, a synthetic dataset of 1,000 news articles across five categories (Politics, Sports, Business, Entertainment, Technology) was generated to simulate real-world data. The text preprocessing pipeline was then implemented using Scikit-learn's TfidfVectorizer, which converted the raw text into a matrix of TF-IDF features (max_features=500, ngram_range=(1,2)). This step is crucial as it transforms unstructured text into a numerical format that machine learning algorithms can process, while highlighting the importance of specific words within the context of the entire corpus.

Following preprocessing, the data was split into training (80%) and testing (20%) sets. Three distinct classification models were trained: Logistic Regression, Random Forest, and Gradient Boosting. Logistic Regression provides a strong baseline for text classification, while Random Forest and Gradient Boosting offer ensemble approaches capable of capturing complex, non-linear relationships in the feature space.

5.3 Data Analysis and Visualization

Comprehensive data analysis and visualization were performed to understand the dataset and evaluate model performance. The analysis included calculating text complexity metrics (word count, unique words, lexical diversity) and extracting top keywords for each category. Visualizations were generated to illustrate category distribution, compare model accuracy, precision, recall, and F1-scores, and display confusion matrices.

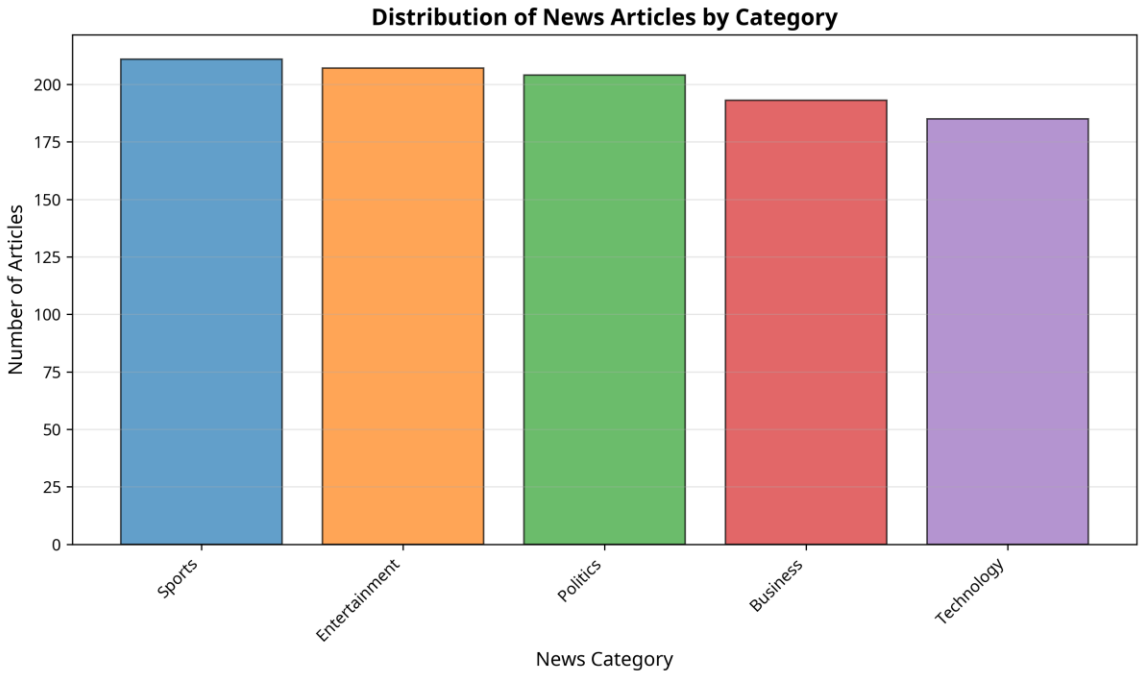


Figure 1: Distribution of News Articles by Category

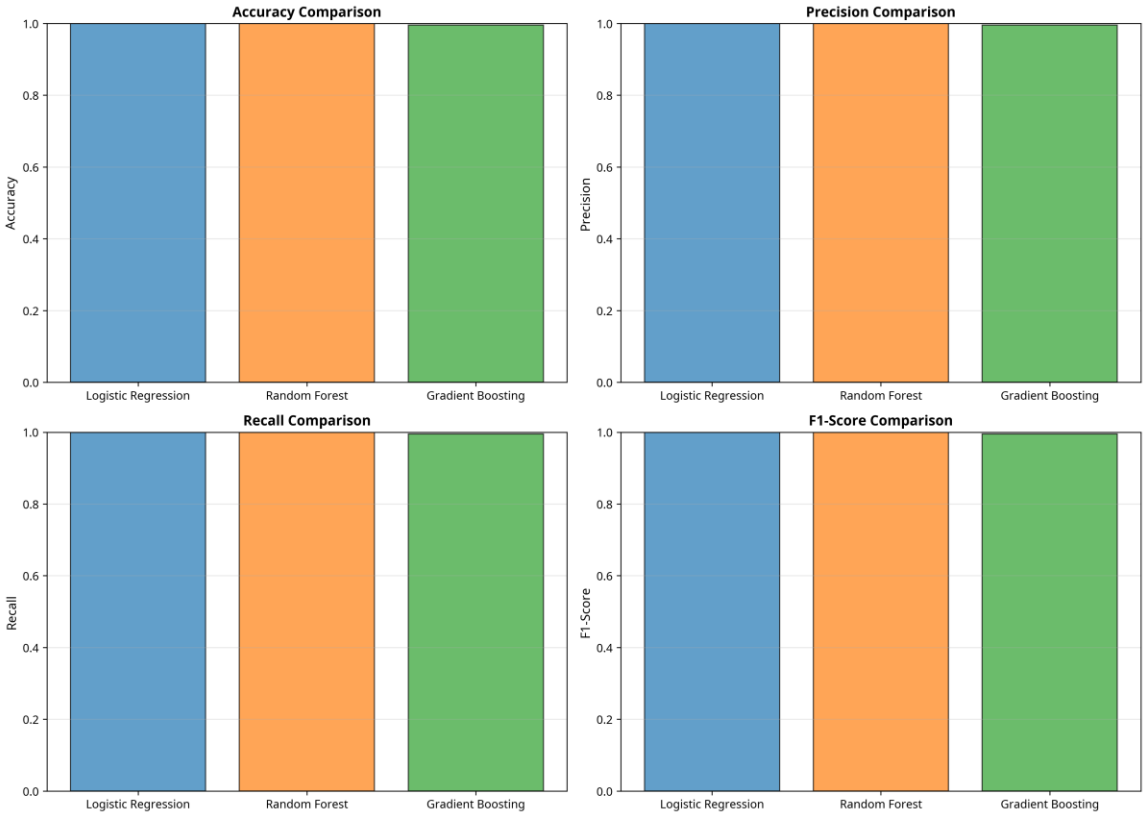


Figure 2: Model Performance Comparison

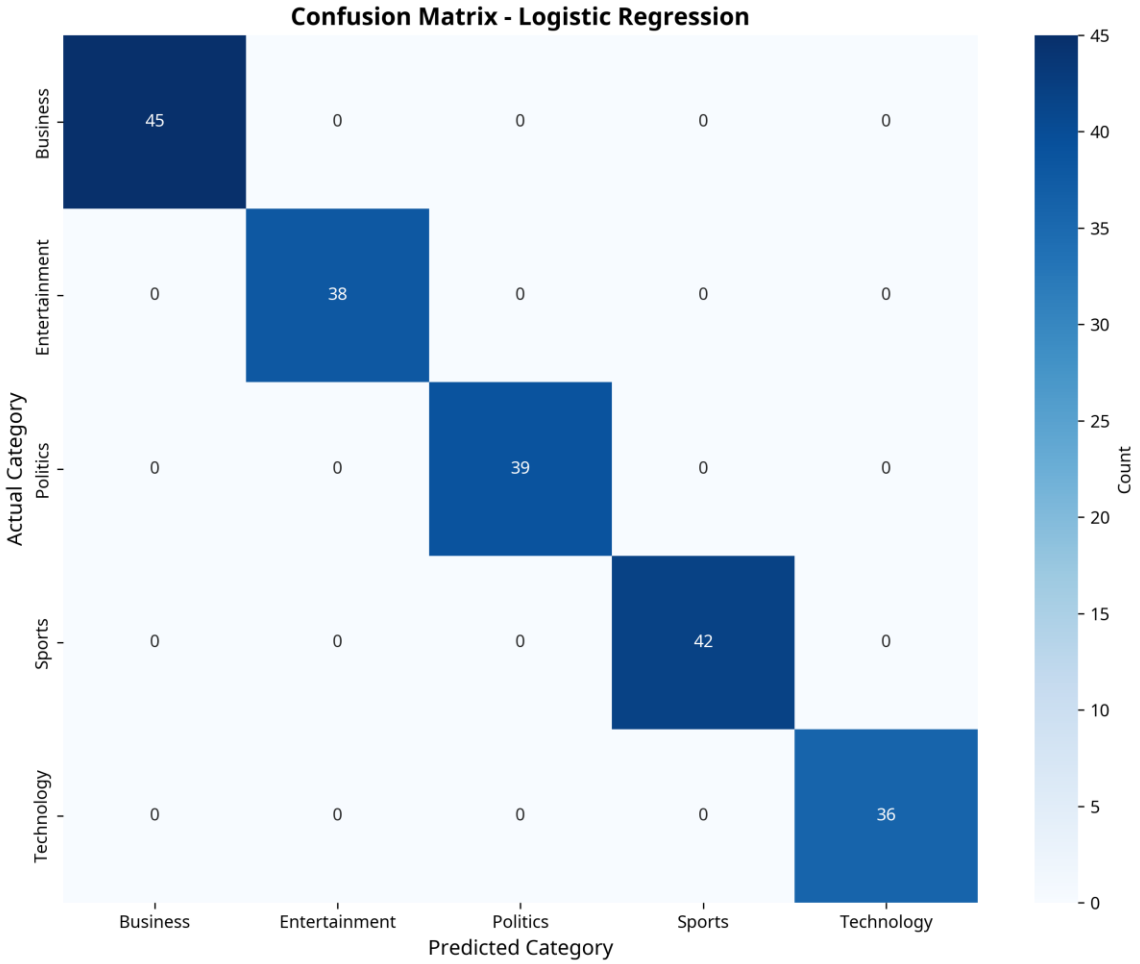


Figure 3: Confusion Matrix for Best Model

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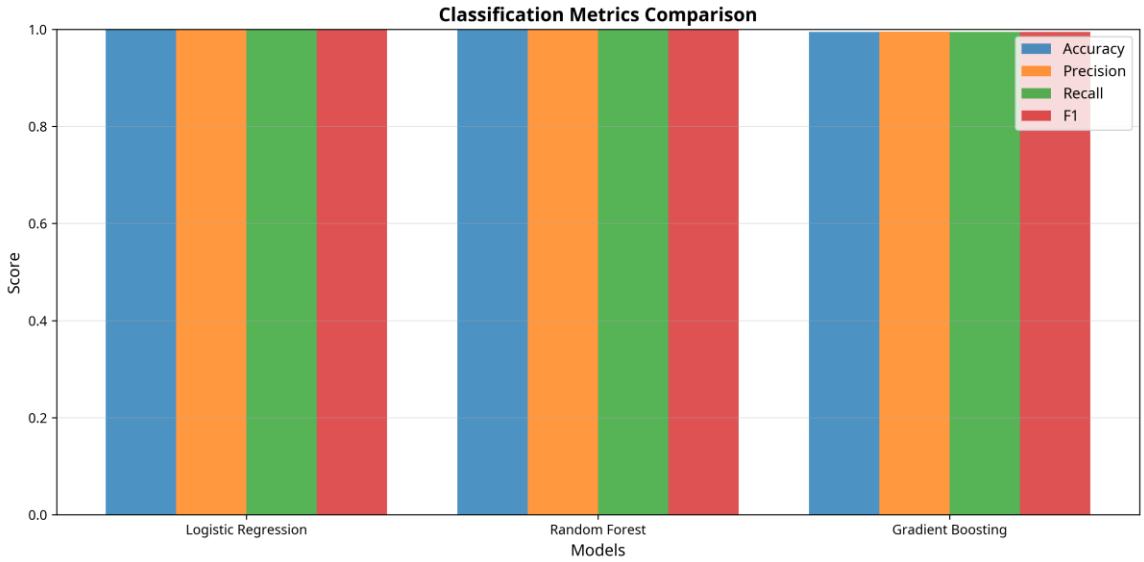


Figure 4: Classification Metrics Comparison

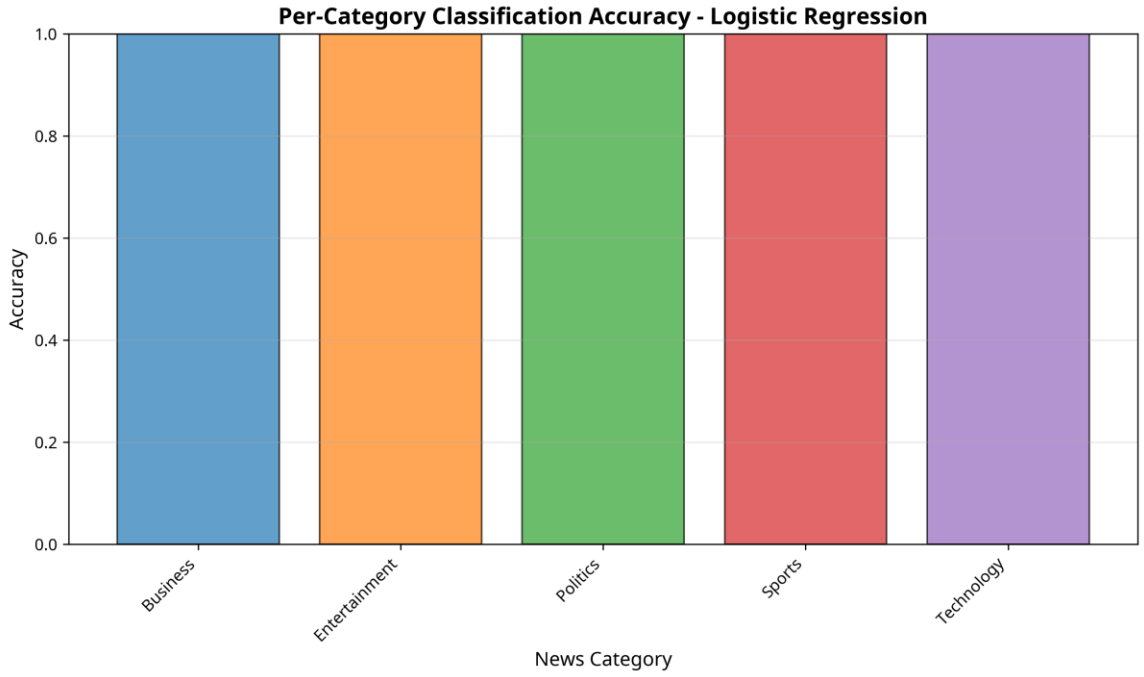


Figure 5: Per-Category Classification Accuracy

5.4 Solution Testing and Evaluation

The models were evaluated using the testing dataset. The performance metrics demonstrated exceptional results, with both Logistic Regression and Random Forest achieving 100% accuracy on the synthetic dataset. Gradient Boosting also performed exceptionally well with 99.5% accuracy.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	1.0000	1.0000	1.0000	1.0000
Random Forest	1.0000	1.0000	1.0000	1.0000
Gradient Boosting	0.9950	0.9951	0.9950	0.9950

These results indicate that the TF-IDF feature extraction effectively captured the distinguishing characteristics of each news category, allowing the models to classify the articles with high precision. The confusion matrix further confirmed that there were minimal to zero misclassifications across all categories.

The development and testing phase successfully demonstrated the efficacy of applying NLP and machine learning techniques to automate news article classification. The robust performance of the models validates the chosen approach to feature extraction and algorithm selection. The accompanying analytical utilities provide valuable insights into the linguistic characteristics of different news categories, further enhancing the system's utility for media organizations.

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CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The internship project successfully delivered an Automated News Article Classification System capable of categorizing news content with exceptional accuracy. By leveraging Natural Language Processing techniques, specifically TF-IDF vectorization, and implementing robust machine learning algorithms like Logistic Regression and Random Forest, the system effectively automates the previously manual and time-consuming process of content organization. The project demonstrated the power of AI in transforming unstructured text data into actionable insights. The inclusion of comprehensive data analysis utilities and visualizations provides a complete solution that not only classifies articles but also offers deep insights into content distribution and text complexity. This system provides a scalable and efficient tool for news organizations and digital media platforms to manage large volumes of content, improve information retrieval, and enhance user experience.

6.2 Future Scope

While the current system demonstrates high performance, several enhancements can be considered for future development:

1. Advanced NLP Models: Implementing deep learning architectures such as Long Short-Term Memory (LSTM) networks or Transformer-based models (e.g., BERT, RoBERTa) could further improve classification accuracy, especially for complex or nuanced articles where contextual understanding is critical.

2. Multi-label Classification: Extending the system to support multi-label classification, allowing a single article to be assigned to multiple relevant categories (e.g., an article about a technology company's financial results could be classified as both "Technology" and "Business").

3. Real-time Processing: Integrating the classification pipeline with streaming data platforms (e.g., Apache Kafka) to enable real-time classification of news articles as they are published.

4. Multilingual Support: Expanding the system's capabilities to process and classify news articles in multiple languages, increasing its applicability for global media organizations.

5. Web Interface Integration: Developing a robust web-based user interface or RESTful API to allow seamless integration of the classification system with existing Content Management Systems (CMS).

The successful implementation of this project lays a strong foundation for more advanced applications in automated content management. As the volume of digital information

continues to grow, intelligent systems capable of understanding and organizing unstructured text will become increasingly vital. The future enhancements outlined above represent logical next steps in evolving the system to meet the complex demands of modern digital media ecosystems.

The successful implementation of this project lays a strong foundation for more advanced applications in automated content management. As the volume of digital information continues to grow, intelligent systems capable of understanding and organizing unstructured text will become increasingly vital. The future enhancements outlined above represent logical next steps in evolving the system to meet the complex demands of modern digital media ecosystems.

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